higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MASONRY
NQF LEVEL 4

22 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1: GENERAL

1.1 D
1.2 B
1.3 A
1.4 C
1.5 D
1.6 D
1.7 A
1.8 B
1.9 B
1.10 C

(10 × 1) **[10]**

QUESTION 2: PLASTERING

2.1 2.1.1 For preparing a smooth off-shutter concrete surface✓ particularly an overhead surface such as soffit.✓ (2)

2.1.2 Using a block brush, forcibly throw it onto the wall in a in a flicking motion.✓ This coat should cover the whole substrate to form a rough texture with nodules about 5mm high.✓ Keep wetting the spatter dash layer so that it does not dry out for at least three days.✓ Before plastering, test its strength and adhesion by scraping it with a trowel.✓ (4)

2.1.3 So that it does not dry out too quickly to rely on hydration for a bond. (1)

2.1.4 Spray a fine mist of water on the substrate.✓ If the water pressure is too strong, you may dislodge the mortar from between the layers of bricks.✓ Ensure that you spray the whole substrate area evenly.✓ Continue spraying until the water is no longer absorbed by the substrate.✓ Once there is uniform run-off from the substrate, the pre-soaking is complete.✓ Leave the area to surface dry overnight and plaster it the next day.✓ Plaster the whole wall in one day to ensure of an even finish.✓ (7)

2.2 Off-shutter concrete is the resultant structure✓ when the shuttering or framework is removed.✓ (2)

2.3 2.3.1 One can use a wire brush to scrub the off-shutter concrete to roughen it enough for the plaster to stick to it.

2.3.2 To check that the surface is level and even.

2.3.3 For cleaning bricks or for giving a rough finish to plaster.

2.3.4 To hold small amounts of mortar to be trowelled up and used to plaster brickwork (masonry).

2.3.5 Is a simple cone-shaped weight that is attached to builder's line and allowed to hang vertically to see whether a wall or any other vertical surface is straight and upright.

(5 × 1) (5)

2.4 A wall chase is a groove, square or rectangular recess✓ cut into a masonry wall✓ to accommodate pipes, heating ducts,✓ and other similar equipment so that these are concealed when the wall is 'made good'.✓ (4)

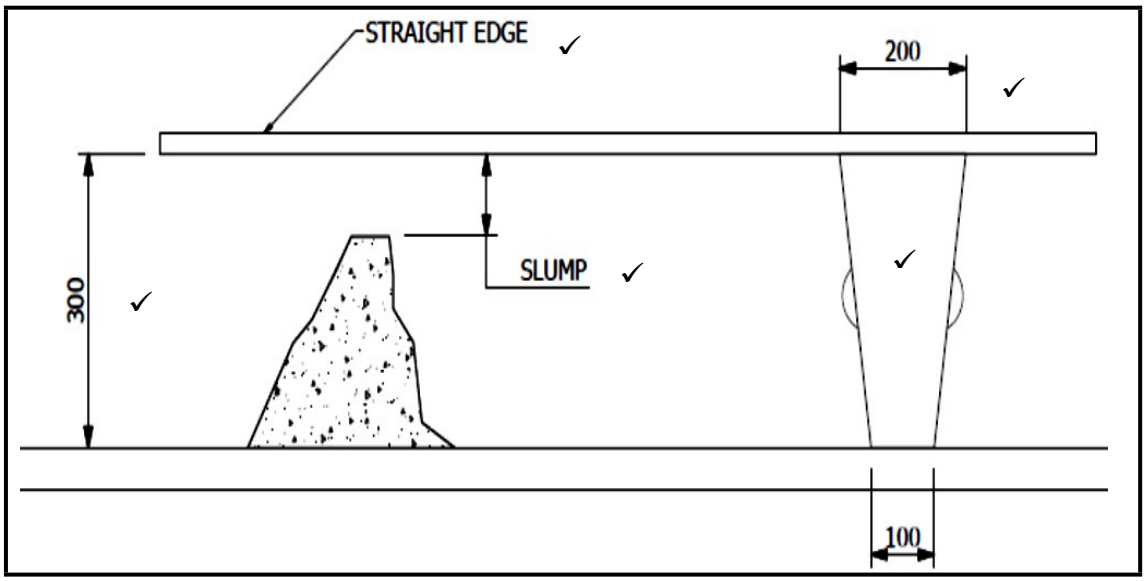
2.5 Measure the plaster sand and spread it on a hard surface.✓ Measure the cement✓ and spread it evenly over the sand.✓Mix thoroughly until the mix attains a uniform grey colour.✓ Form a dam in the mix and add water.✓ Mix the ingredients well.✓ (Any 5 × 1) (5)

[30]

QUESTION 3: SCREED AND TOPPINGS

3.1 The datum level or site datum is the fixed level✓ on the building site or construction project.✓ Where a concrete slab can be finished off to the line✓ and datum level in compliance with the contract specification, a screed or topping may be optional (not necessary).✓ (4)

3.2 Hold the slump mould in position and fill it with concrete in four equal layers, rodding (poking) each layer exactly 25 times. ✓ Strike off surplus concrete from the cone and clear droppings around the base.✓ Lift the cone off steadily, vertically and place the cone alongside the slumped concrete.✓ Place a straight edge across the cone and measure the difference in height.✓ The slump is now measured and recorded as a collapse or shear slump. ✓



(10)

3.3
- Repair gaps or cracks in the substrate.
- Clean the substrate.
- Create a mechanical key to ensure that a strong bond is formed between the substrate and the screed or topping. (3)

3.4 Clean the substrate by vacuum cleaning✓ and using oil-free compressed air,✓ sweeping dust and dirt,✓ chiselling off the bits of concrete or old screed,✓ brushing with a wire brush.✓ (Any 2 × 2) (2)

3.5
- Work a small area at a time.
- Slush the grout mix over the area where it is needed.
- Brush the grout in manually, using a stiff nylon scrubbing brush.
- Use a brush with bristles of 30–50 mm long, so that they are flexible enough to reach into any depressions. Brush the grout out again, leaving only a very thin coating on the surface with no pools of grouting in the depressions.
- Lay the screed within 10 to 20 minutes. (5)

3.6 Use a strong detergent✓ and boiling water to help remove grease.✓ (2)

3.7
- Static weight rollers.
- Vibratory rollers.
- Pneumatic rollers.
- Mini vibrating rollers.
- Vibrating rollers.
- Impact plates. (Any 4 × 1) (4)

[30]

QUESTION 4: TILING

4.1 Tile adhesive hardens to the same degree and strength as cement mixtures.✓ The mixture will cure and harden in the piping, especially in traps.✓ The tile adhesive in the outlet system will create a full or partial blockage.✓ The drainage outlet system is under the flooring and tiles.✓ A blockage discovered after the system of outlets and drains has been connected will entail reopening the drainage system✓ and the replacing of the blocked pipes.✓ This will be a very costly operation and may well interfere with the overall aesthetics of the completed area.✓ (Any 6 × 1) (6)

4.2 Low-tech testing:✓
- Visual inspection is the primary tool for inspecting plasterwork.✓
- A simply visual inspection can often reveal cracks, popping, grinning or other defects.✓
- Touching the surface can reveal loose plaster and areas of moisture.✓
- Application of a little pressure can reveal detachment by the springiness of the surface.✓
- Tapping the plaster, the trained ear can sense the areas that are hollow sounding. (5 × 1)

High-tech testing:✓

- The oldest method is the electrical resistance damp meter.✓
- Infrared thermography that involves heating the surface of the plaster and recording the result on an infrared video camera.✓
- Electronic speckle pattern interferometry (ESPI) in which micro-deformations are monitored.✓
- Laser scanning vibrometry is a non-invasive, non-destructive method that measures the 3D micro deformations of the surface plaster.✓ (5 × 1) (10)

4.3 As each new layer of the plastered surface must bond adequately to the substrate below it, so to, the tiles must bond to the plastered surface adequately.✓ Tile adhesive does not bond adequately to painted surfaces.✓ Old, flaky, high-gloss,✓ or smooth finishes do not allow for bonding.✓The surface must be properly prepared by stripping down to the underlying plaster to give a stippled effect.✓ To provide a good bond, this stippling need not be too deep but must cover at least 80% of the surface to be tiled.✓
The condition and strength of the surface to be tiled plays the biggest part in the performance and success of any tiling system.✓
The preparation of the surfaces is therefore of the utmost importance.✓
The surface to be tiled must be capable of supporting the weight of the tiles without breaking down.✓ All surfaces must be firm, free of dust, mould, oil, grease, wax polish and organic growth.✓ (10 × 1) (10)

4.4 4.4.1 plastic sheeting
4.4.2 seven days
4.4.3 21 days
4.4.4 28 days

(4 × 1) (4)
[30]

TOTAL: 100